	B. M.S. COLLEGE OF ENGINEERING, BANGALORE-19 (Autonomous Institute, Affiliated to VTU) **Department Name: CSE**

First INTERNALS – Online		
Course Code : 19CS4PCTFC	**Course Title :** Theoretical Foundations of Computations	
Semester : 4th	**Maximum Marks: 40**	**Date: 17-5-2021**
Faculty Handling the Course:	JSN, GS, UV, GRP	
Instructions: ***Internal choice is provided in Part C.***		

PART-A

Total 5 Marks (No choice)

No.	Question	Marks	CO No.	Level
1a	**i.** Explain the operators used in Regular Expressions along with their precedence. **ii.** Compute L* given L={0,11}	**5M**	**1**	**1**

PART-B

Total 15 Marks (No Choice)

No.	Question	Marks	CO No.	Level
2a	Obtain DFA for accepting binary strings which begin with 101	**5M**	**1, 2**	**2**
2b	Convert the following NFA to DFA (table below)	**5M**	**1, 2**	**2**
2c	Given the following ε-NFA obtain its equivalent DFA using ε-closure. (table below)	**5M**	**1, 2**	**2**

2b — NFA table:

	a	b
→p	{p,q}	{p}
q	ϕ	{r}
*r	{p,r}	{q}

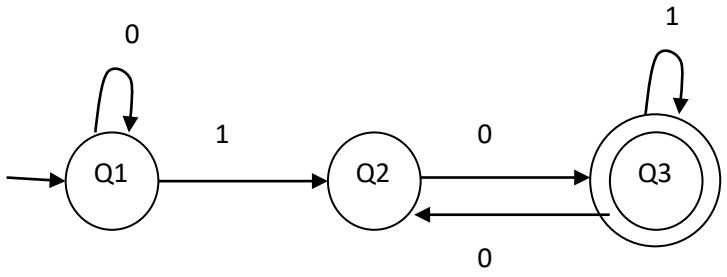
2c — ε-NFA table:

	0	1	ε
→q0	{q1,q2}	{q0}	{q1}
q1	ϕ	{ q1}	{q2}
*q2	{q2}	{q2}	ϕ

PART- C

Total 20 Marks (Choice)

No.	Question	Marks	CO No.	Level

3a	Design DFA for the following over the alphabet set $\sum = \{a,b\}$ **i.** all strings over the alphabet that do not contain the substring 'bba'. **ii.** The set of all strings except those which end with 'abb' **iii.** The set of all strings that contains consecutive three a's **iv.** The set of all strings of length at least two.	**10M**	**1,3**	**3**
	OR			
3b	**I.** Design Regular Expressions (RE) for the following language over the alphabet set $\sum = \{a, b\}$ **i.** Strings of a's and b's containing no more than three b's **ii.** Strings of a's and b's ending with either "aba" or "aaba" **iii.** Strings of a's and b's such that fourth symbol from right end is "b" and fifth symbol from right end is "a" **iv.** $L = \{a^n b^m, (n+m) \text{ is even}\}$ **II.** Design ε-NFA for the RE (a+b)*ab*	**10M**	**1,3**	**3**
4a	Show that the following languages are not regular using Pumping lemma. **i.** $L=\{1^n, n \text{ is prime}\}$ **ii.** $L = ww^R$, $w \in (a+b)^*$	**10M**	**1,3**	**3**
	OR			
4b	Convert the following DFA to a RE, using the Kleene's technique.  	**10M**	**1, 3**	**3**

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First INTERNALS – Online		
Course Code : 19CS4PCTFC	**Course Title :** Computer Organization and Architecture	
Semester : 4th	**Maximum Marks: 40**	**Date: 17-5-2021**
Faculty Handling the Course:	JSN, GS, UV, GRP	
Instructions: ***Internal choice is provided in Part C.***		

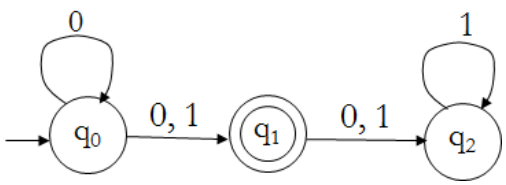
PART-A

Total 5 Marks (No choice)

No.	Question	Marks	CO No.	Level
1a	**i.** Write the formal definition of DFA **ii.** List any two differences between Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NFA)	**5M**	**1**	**1**

PART-B

Total 15 Marks (No Choice)

No.	Question	Marks	CO No.	Level
2a	Obtain NFA for the following i. NFA that accepts the language L(aa*(a+b)) ii. NFA that accepts the language L={a^3 ∪ a^{2n}, n>=1}	**5M**	**1, 2**	**2**
2b	Convert the following NFA to DFA  	**5M**	**1, 2**	**2**
2c	Given the following ε-NFA obtain its equivalent DFA using ε-closure. | | ε | a | b | c | |---|---|---|---|---| | →p | {q,r} | ϕ | {q} | {r} | | q | ϕ | {p} | {r} | {p,q} | | *r | ϕ | ϕ | ϕ | ϕ |	**5M**	**1, 2**	**2**

PART- C

Total 20 Marks (Choice)

No.	Question	Marks	CO No.	Level
3a	Design DFA for the following over the alphabet set ∑ = {a, b} **i.** Not more than three a's in the strings **ii.** The set of all strings except those which end with 'abb'	**10 M**	**1,3**	**3**

	iii. The set of all strings either begins or ends or both with substring 'ab' **iv.** L={w / n_a(w) mod 3 ==0 }			
	OR			
3b	**I.** Design Regular Expressions (RE) for the following language over the alphabet set ∑ ={a, b} **i.** Strings of a's and b's containing no more than three a's **ii.** Strings of a's and b's ending with either "ab" or "ba" **iii.** Strings of a's and b's such that fourth symbol from right end is "a" and fifth symbol from right end is "b" **iv.** L = {$a^n b^m$, n>=1, m>=1, nm>=3} **II.** Design ε-NFA for the RE (0+1)*1(0+1)	**10 M**	**1,3**	**3**
4a	Show that the following languages are not regular using Pumping lemma. **i.** L={ww , w ∈ (a+b)*} **ii.** L = {$a^{n!}$, n>=0}	**10 M**	**1,3**	**3**
	OR			
4b	Convert the following DFA to a RE, using the Kleene's technique. | | 0 | 1 | |→p | q | p | |*q | q | q |	**10 M**	**1,**	**3**

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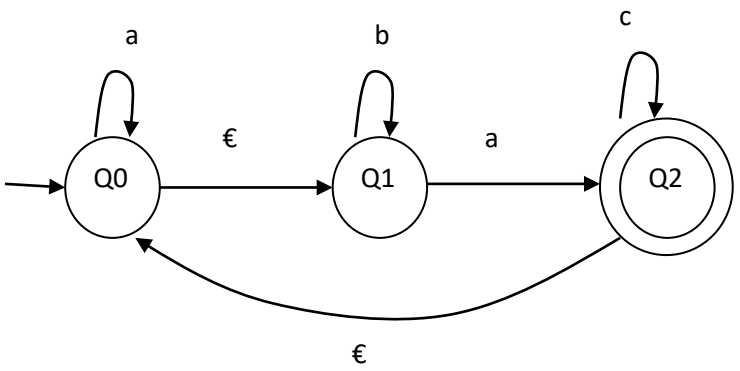
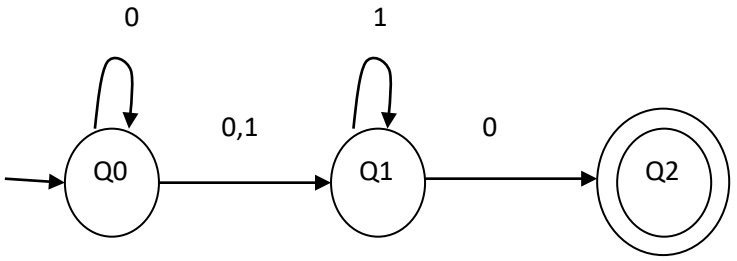
PART-A

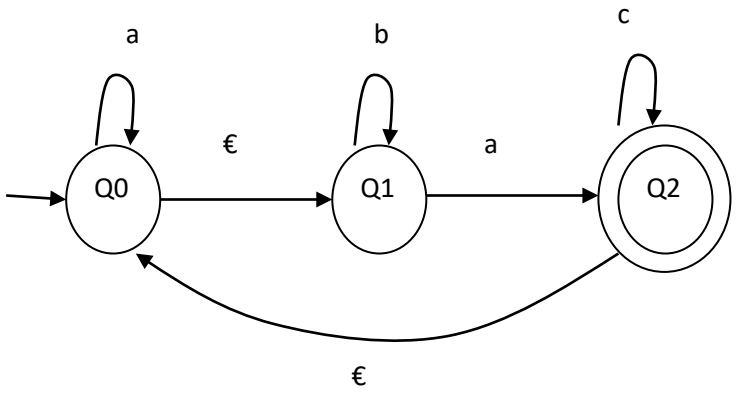
Total 5 Marks (No choice)

No.	Question	Marks	CO No.	Level
1a	**i.** Define epsilon closure of state. **ii.** Find epsilon closure of each state in the bellow diagram  	**5M**	**1**	**1**

PART-B

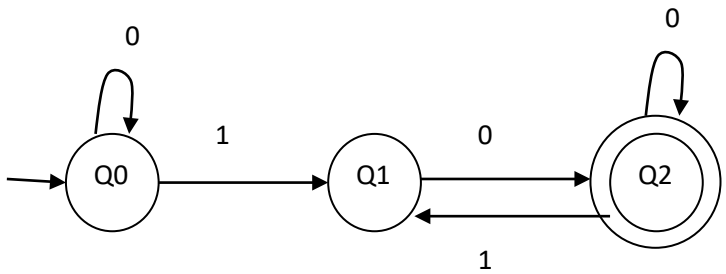
Total 15 Marks (No Choice)

No.	Question	Marks	CO No.	Level
2a	Obtain DFA for accepting binary strings whose value is divisible by 3	**5M**	**1, 2**	**2**
2b	Convert the following NFA to DFA  	**5M**	**1, 2**	**2**

2c	Given the following ε-NFA obtain its equivalent DFA. 	**5M**	**1, 2**	**2**

PART- C

Total 20 Marks (Choice)

No.	Question	Marks	CO No.	Level
3a	Design DFA, over the alphabet set ∑ = {a, b}, for accepting the strings **i.** Having exactly 3 a's **ii.** Not having the pattern 'aab' **iii.** Whose length is multiple of 3 **iv.** Start with 'aab' and string are of even length.	**10 M**	**1,3**	**3**
	OR			
3b	**I.** Design Regular Expressions (RE) for the following language over the alphabet set ∑ ={a, b} **i.** Strings having alternate a's and b's **ii.** L={$a^n b^{m+2}$| n>0,m>=0} **iii.** Strings of a's and b's having odd number of a's. **II.** Design ε-NFA for the RE 1(01*)*(0+1)	**10 M**	**1,3**	**3**
4a	Show that the following languages are not regular using Pumping lemma. **i.** L={1^p| p is a perfect square} **ii.** L = {$0^n 1 0^n$| n>1}	**10 M**	**1,3**	**3**
	OR			
4b	Convert the following DFA to RE, using the Kleene's technique. 	**10 M**	**1,**	**3**